

Core Module Testing Tools

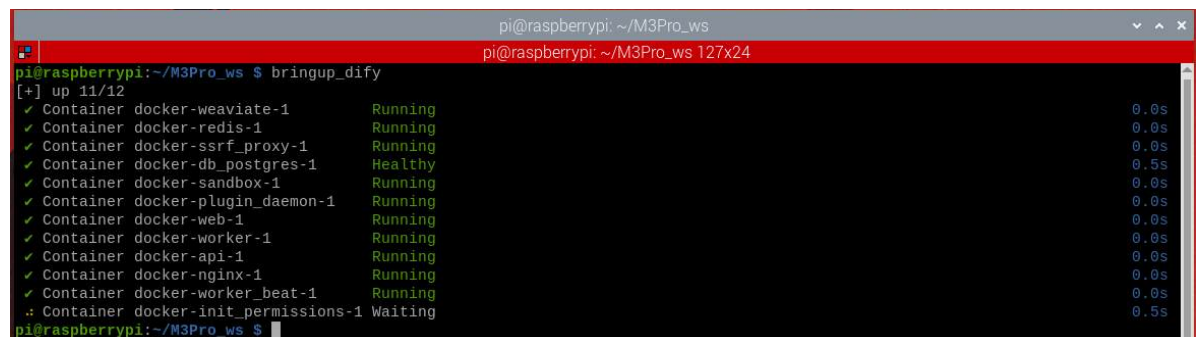
Core Module Testing Tools

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1. Course Content

- multi_brains provides minimal test programs for testing the core modules of a large model, used to quickly locate problems in abnormal situations.

2. Start the Dify service



```
pi@raspberrypi: ~/M3Pro_ws
pi@raspberrypi: ~/M3Pro_ws 127x24
pi@raspberrypi:~/M3Pro_ws $ bringup_dify
[+] up 11/12
✓ Container docker-weaviate-1      Running      0.0s
✓ Container docker-redis-1        Running      0.0s
✓ Container docker-ssrf_proxy-1    Running      0.0s
✓ Container docker-db_postgres-1   Healthy      0.5s
✓ Container docker-sandbox-1       Running      0.0s
✓ Container docker-plugin_daemon-1 Running      0.0s
✓ Container docker-web-1           Running      0.0s
✓ Container docker-worker-1        Running      0.0s
✓ Container docker-api-1           Running      0.0s
✓ Container docker-nginx-1         Running      0.0s
✓ Container docker-worker_beat-1    Running      0.0s
✗ Container docker-init_permissions-1 Waiting      0.5s
pi@raspberrypi:~/M3Pro_ws $
```

3. Navigate to the test tool path

```
cd ~/M3Pro_ws/src/multi_brains/test
ls
```

- Test tools are as follows:



```
test_PiperTTS.py      test_TongyiASR.py    test_copyright.py    test_flake8.py      test_xunfeiASR.py
test_SenseVoiceSmall.py test_TongyiTTS.py    test_dify_connection.py test_pep257.py      test_xunfeiTTS.py
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

4. Test local speech synthesis function

- Test the local speech synthesis function for Chinese and English in sequence

```
python3 test_PiperTTS.py
```

After running, it will play the Chinese and English speech synthesis results sequentially.

```
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_PiperTTS.py
pygame 2.6.1 (SDL 2.28.4, Python 3.10.12)
Hello from the pygame community. https://www.pygame.org/contribute.html
测试中文语音合成
Test English speech synthesis
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

5. Test Local Speech Recognition Functionality

- After running, it will test the preset Chinese and English audio recordings sequentially.

```
python3 test_SenseVoiceSmall.py
```

After running, it will print the speech recognition results of the test audio.

```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test 127x24
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_SenseVoiceSmall.py
=====测试本地中文语音识别=====
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf_avg: 0.463: 100%|██████████████████████████████████████████████████████████████████████████| 1/1 [00:01<00:00, 1.36s/it]
你好我是巨身智能机器人
=====Test local English speech recognition=====
funasr version: 1.2.6.
SenseVoiceSmall ASR_Engine initialized successfully
rtf_avg: 0.353: 100%|██████████████████████████████████████████████████████████████████████████| 1/1 [00:01<00:00, 1.21s/it]
hello i'm embodied intelligent robot
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

6. Testing Whether the Robot Can Access Dify Normally

- If the program indicates that the model service is unavailable—and you are unsure whether this is due to an incorrect network address preventing the robot/vehicle system from accessing Dify—you can perform the following test:

```
python3 test_dify_connection.py
```

If the robot successfully displays the information shown below, it confirms that the connection to Dify is functioning correctly. Otherwise, Dify may not have started, or the Dify program may have crashed and requires a restart.

```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test 127x24
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_dify_connection.py
/root/M3Pro_ws/multi_brains_file/multi_brains_setting.yaml
Connection dify successful
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

7. Online Voice Services for Users in Mainland China

7.1 Baillian Speech Recognition (ASR)

- If errors occur with the online speech recognition service, you can independently test the Baillian speech recognition service:

```
python3 test_TongyiASR.py
```

- Upon execution, the script will perform a speech recognition test using a preset audio sample.

```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test 127x24
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_TongyiASR.py
ASR init success
你好，我是巨深智能机器人。
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

7.2 Baillian Speech Synthesis (TTS)

- If errors occur with the online speech synthesis service, you can independently test the Baillian speech synthesis service:

```
python3 test_TongyiTTS.py
```

Upon execution, the script will first synthesize speech from a default text sample, and then play the corresponding audio.

```
root@raspberrypi: ~/M3Pro_ws/src/multi_br
root@raspberrypi: ~/M3Pro_ws/src/multi_brai
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# python3 test_TongyiTTS.py
pygame 2.6.1 (SDL 2.28.4, Python 3.10.12)
Hello from the pygame community. https://www.pygame.org/contribute.html
Tongyi TTS_Engine initialized successfully
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test# █
```

7.3 Baidu Qianfan Speech Synthesis (TTS)

- If errors occur while using Baidu's speech synthesis service, you can independently test the Baidu speech synthesis service:

```
python3 test_baiduTTS.py
```

Upon execution, the script will first synthesize speech from a default text sample, and then play the corresponding audio.

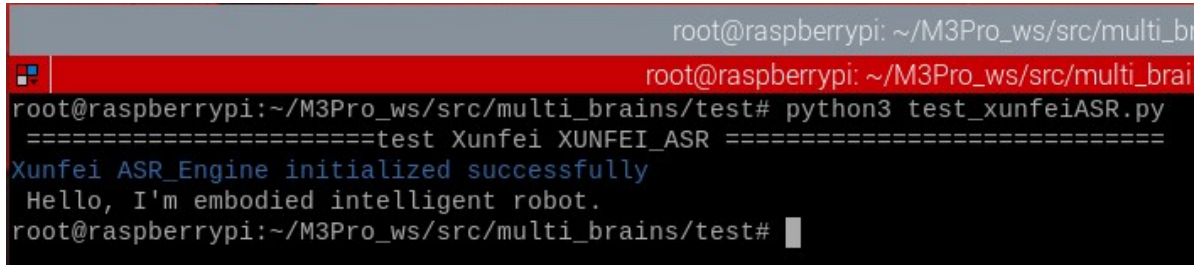
8. Online Voice Services for International Users

8.1 iFLYTEK Speech Recognition Service

- If international users encounter errors with online speech recognition, they can test the iFLYTEK speech recognition service independently.

```
python3 test_xunfeiASR.py
```

- Upon execution, the script will perform a speech recognition test on a pre-configured test audio file.

A terminal window screenshot showing the execution of a Python script. The prompt is root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test. The command executed is python3 test_xunfeiASR.py. The output shows a series of equals signs, followed by 'Xunfei XUNFEI_ASR', then 'Xunfei ASR_Engine initialized successfully', and finally 'Hello, I'm embodied intelligent robot.' The prompt returns to root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test#.

```
root@raspberrypi: ~/M3Pro_ws/src/multi_brains/test# python3 test_xunfeiASR.py
=====test Xunfei XUNFEI_ASR =====
Xunfei ASR_Engine initialized successfully
Hello, I'm embodied intelligent robot.
root@raspberrypi:~/M3Pro_ws/src/multi_brains/test#
```

8.2 iFLYTEK Speech Synthesis Service

- If international users encounter errors with online speech synthesis, they can test the iFLYTEK speech synthesis service independently.

```
python3 test_xunfeiTTS.py
```

Upon execution, the script will first synthesize speech from a default text sample, and then play the corresponding audio.